

SGA 3, Exposé XIII
Regular Elements of Algebraic Groups
and Lie Algebras
English Reconstruction

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Contents

Exposé XIII. Regular Elements of Algebraic Groups and Lie Algebras	1
1 An auxiliary lemma on varieties with operators	1
2 Density theorem and theory of the regular points of G	4
3 The case of an arbitrary base prescheme	14
4 Lie algebras over a field: rank, regular elements, Cartan subalgebras	18
5 The case of the Lie algebra of a smooth algebraic group: the density theorem	23
6 Cartan subalgebras and subgroups of type (C) associated with a smooth algebraic group	27

Exposé XIII

Regular Elements of Algebraic Groups and Lie Algebras

by A. Grothendieck

1 An auxiliary lemma on varieties with operators

Let S be a prescheme, let G be an S -group prescheme acting on the left on an S -prescheme V , let W be a closed S -subprescheme of V , and let N be its stabilizer in G . Thus N is the subgroup of G whose points with values in an S' over S are the elements $g \in G(S')$ such that

$$g \cdot W_{S'} = W_{S'}.$$

We endow (Sch/S) with the faithfully flat quasi-compact topology and identify G , V , and W with the corresponding sheaves.¹ We shall therefore reason in the category of sheaves on (Sch/S) ; throughout this section, “locally” refers to the topology just specified on (Sch/S) .

Since N is a sheaf, consider the quotient sheaf G/N . It is immediately seen to be isomorphic to the following functor: to every S' over S , it assigns the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G . Let X be the subsheaf of $G/N \times_S V$ whose value on an S' over S is the set of pairs (W', v) , where W' is as above and v is a section of W' over S' (hence also a section of $V_{S'}$ over S'). Let Z be the inverse image of X in $G \times_S V$. We then have the cartesian diagram

$$(x) \quad \begin{array}{ccc} Z & \longrightarrow & X \\ \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V, \end{array}$$

where i is the canonical immersion, while the lower horizontal arrow is induced by the canonical morphism $G \rightarrow G/N$.

This morphism sends $g \in G(S')$ to the subsheaf $g \cdot W_{S'}$ of $V_{S'}$. Consequently, $Z(S')$ is the set of pairs $(g, v) \in G(S') \times V(S')$ such that $v \in (g \cdot W_{S'})(S')$. It follows that Z is isomorphic to the sheaf $G \times_S W$, via the isomorphism

$$G \times_S W \xrightarrow{\sim} Z, \quad (g, w) \longmapsto (g, g \cdot w).$$

¹The French source prints M in this list; the surrounding sentence and all subsequent uses require W .

Thus the preceding cartesian diagram yields the cartesian diagram

$$(xx) \quad \begin{array}{ccc} G \times_S W & \xrightarrow{q} & X \\ \lambda \downarrow & & \downarrow i \\ G \times_S V & \longrightarrow & G/N \times_S V \end{array}$$

where

$$\lambda(g, w) = (g, g \cdot w), \quad q(g, w) = (\bar{g}, g \cdot w),$$

where $\bar{g}$ denotes the image of g under the canonical map $G(S') \rightarrow (G/N)(S')$.

Diagram (x) also shows that $Z \rightarrow X$ makes Z a principal bundle over X , with N acting on the right by

$$(g, v) \cdot n = (gn, v).$$

Accordingly, in diagram (xx), the morphism $q : G \times_S W \rightarrow X$ makes $G \times_S W$ a principal bundle over X , with right N -action

$$(g, w) \cdot n = (gn, n^{-1} \cdot w).$$

The principal morphisms just introduced are summarized by

$$(D) \quad \begin{array}{ccccc} & & G \times_S W & & \\ & & \downarrow q & \searrow \varphi & \\ G/N & \xleftarrow{p} & X & \xrightarrow{\psi} & V \\ & \swarrow \text{pr}_1 & \downarrow i & \searrow \text{pr}_2 & \\ & & G/N \times_S V & & \end{array}$$

where

$$\psi = \text{pr}_2 \circ i, \quad \varphi = \psi \circ q, \quad \varphi(g, w) = g \cdot w.$$

If v is a section of V over S , let X_v be its inverse image in X under ψ . For every S' over S , $X_v(S')$ is the set of subsheaves W' of $V_{S'}$ that are locally conjugate to $W_{S'}$ under G and contain the section $v_{S'}$ of $V_{S'}$. The inverse image of v in $G \times_S W$ under φ is isomorphic to the subsheaf M_v of G defined by

$$M_v(S') = \{g \in G(S') \mid v_{S'} \in (g \cdot W_{S'})(S')\} = \{g \in G(S') \mid g^{-1}v_{S'} \in W(S')\}.$$

If v is a section of W , and not merely of V , then M_v evidently contains N .

None of these explicit descriptions used the representability of G , V , or W (nor even the fact that the site in question is defined in terms of preschemes). Suppose now, however, that N is representable, faithfully flat, and quasi-compact over S , and that G/N is representable. When S is the spectrum of a field and G is of finite type over that field, this hypothesis is necessarily satisfied (VIII 6 and VI_B.11.18). Using the cartesian diagram (xx), faithfully flat quasi-compact descent, and the fact that $Z \rightarrow G \times_S V$ is a closed immersion, one sees that X is representable: it is obtained by descending the closed subprescheme Z of $G \times_S V$ along the faithfully flat quasi-compact morphism

$$G \times_S V \longrightarrow G/N \times_S V.$$

Thus diagram (D) is a diagram of morphisms of preschemes over S .

We henceforth assume that S is the spectrum of a field k , and that G , V , and W are of finite type over k . Let $\mathfrak{n}$ be the Lie algebra of N . Then $\dim N \leq \text{rank}_k \mathfrak{n}$, with equality if and only if N is smooth over k (Exposé VI²). Let $a \in W(k)$, and consider the subscheme M_a of G defined above; it contains N and is isomorphic to $\varphi^{-1}(a)$. We denote by $\mathfrak{m}_a$ its Zariski tangent space at the identity element e of N . Thus³

$$(1) \quad \mathfrak{n} \subset \mathfrak{m}_a, \quad \dim N \leq \text{rank}_k \mathfrak{n} \leq \text{rank}_k \mathfrak{m}_a.$$

Lemma 1.1. — *With the preceding notation:*

a) Consider the following conditions:

- (i) $\mathfrak{n} = \mathfrak{m}_a$, and N is smooth over k .
- (i bis) $\dim N = \text{rank}_k \mathfrak{m}_a$.
- (ii) The morphism $\psi : X \rightarrow V$ is unramified at $(\bar{e}, a)$.
- (iii) M_a and N coincide in a neighborhood of e .

Then

$$(i) \iff (i \text{ bis}) \implies (ii) \iff (iii).$$

b) Suppose that $\varphi : G \times_S W \rightarrow V$ is smooth at (e, a) . Then M_a is smooth over k at e , and ψ is smooth at $(\bar{e}, a)$.

Proof. a) The equivalence of (i) and (i bis) follows immediately from (1) and from the fact noted above that N is smooth over k if and only if $\dim N = \text{rank}_k \mathfrak{n}$. Consider the inclusion morphism $N \rightarrow M_a$. It is well known⁴ that if N is smooth over k at e and the tangent map at e is surjective, then $N \rightarrow M_a$ is smooth at e ; since it is an immersion, it is therefore an isomorphism at e . This proves that (i) implies (iii).

To prove the equivalence of (ii) and (iii), let $X_a = \psi^{-1}(a)$, as above. From the isomorphisms

$$M_a \simeq \varphi^{-1}(a) \simeq q^{-1}(X_a)$$

we obtain a morphism $p_a : M_a \rightarrow X_a$ making M_a a principal homogeneous bundle with group N_{X_a} . Consider the diagram

$$\begin{array}{ccc} M_a & \xleftarrow{j_a} & N \\ p_a \downarrow & & \downarrow \\ X_a & \xrightarrow{j'_a} & S = \text{Spec}(k). \end{array}$$

Here $\text{Spec}(k) \rightarrow X_a$ ⁵ is defined by the point $\bar{a} = (\bar{e}, a)$ of X , and $j_a : N \rightarrow M_a$ is the canonical immersion. To say that ψ is unramified at $(\bar{e}, a)$ means that j'_a is an open

²Editor's note in the French source: since this reference could not be identified, see Theorem II.5.2.1 of M. Demazure and P. Gabriel, *Groupes algébriques I*, Masson (1970).

³The French source omits rank_k before $\mathfrak{m}_a$ in (1); it is restored here, as required by the surrounding text and Lemma 1.1.

⁴Editor's note in the French source: see, for example, EGA IV₄, Theorem 17.11.1(d).

⁵The printed diagram directs $j'_a : X_a \rightarrow S$, whereas the immediately following prose defines the point morphism $S \rightarrow X_a$ and calls it j'_a . Both the printed diagram direction and the verbal definition are retained here. The source also prints X_n here and $X_{a,b}$ in the two local rings below. They are corrected to X_a and $X_{a,\bar{a}}$, in accordance with the fiber notation and the point $\bar{a}$.

immersion, or equivalently that it induces an isomorphism

$$S \xrightarrow{\sim} \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}}).$$

Since p_a is flat, this is equivalent to saying that the morphism obtained from the preceding one by the base change

$$\operatorname{Spec}(\mathcal{O}_{M_a, e}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{X_a, \bar{a}})$$

is an isomorphism. The resulting morphism is precisely

$$\operatorname{Spec}(\mathcal{O}_{N, e}) \longrightarrow \operatorname{Spec}(\mathcal{O}_{M_a, e}),$$

which proves the equivalence of (ii) and (iii).

Notice moreover that the proof shows that (ii) and (iii) imply the following condition, apparently stronger than (iii):

(iii bis) N is an open and closed subscheme of M_a .

b) The first assertion follows from the isomorphism $M_a \simeq \varphi^{-1}(a)$, and the second from the fact that q is flat and $q(e, a) = (\bar{e}, a)$.

2 Density theorem and theory of the regular points of G

We apply the constructions and notation of the preceding section in the following situation. Let G be a smooth connected algebraic group over k ; take $V = G$, with G acting on V by inner automorphisms; and let $W = H$, where H is a smooth connected algebraic subgroup of G . We denote the Lie algebras of G and H by $\mathfrak{g}$ and $\mathfrak{h}$, the normalizer of H in G by N , and its Lie algebra by $\mathfrak{n}$. For $a \in G(k)$, we again denote by M_a , as in Section 1, the inverse of the transporter of a into H . Thus, if $a \in H(k)$, then $N \subset M_a$; in that case $\mathfrak{m}_a$ denotes the Zariski tangent space of M_a at the identity element e of G . Note that

$$\mathfrak{h} \subset \mathfrak{n} \subset \mathfrak{m}_a \subset \mathfrak{g} \quad \text{for } a \in H(k).$$

We shall use the following lemma.

Lemma 2.0. — *The equality $H = N^0$ holds if and only if*

$$(\mathfrak{g}/\mathfrak{h})^H = 0,$$

where the left-hand side denotes the subspace of invariants for the action of H induced by the adjoint representation. When this condition holds, N is smooth and $\dim X = \dim G$. In all cases, $\dim X \leq \dim G$, and equality holds if and only if H has finite index in N .

Indeed, by II 5.2.3(i), $\mathfrak{n}$ is the inverse image of $(\mathfrak{g}/\mathfrak{h})^H$ under $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$. Hence $(\mathfrak{g}/\mathfrak{h})^H = 0$ is equivalent to $\mathfrak{h} = \mathfrak{n}$, which in turn is equivalent to $H = N^0$, since H is a smooth connected algebraic subgroup of the algebraic group N (cf. VI.2). This also shows that N is smooth. Furthermore,

$$\dim X = (\dim G - \dim N) + \dim H = \dim G - (\dim N - \dim H),$$

and therefore

$$\dim X \leq \dim G,$$

with equality if and only if $\dim H = \dim N$, that is, if and only if H has finite index in N . This holds in particular when $H = N^0$, and proves the lemma.

Theorem 2.1. — *Let G be a smooth connected algebraic group over an algebraically closed field k , let H be a smooth connected algebraic subgroup, and let N be its normalizer. Write $\mathfrak{g}, \mathfrak{h}, \mathfrak{n}$ for their Lie algebras, let $X = G \times^N H$ be the scheme introduced in Section 1 (fibered over G/N with typical fiber H), and let $\psi : X \rightarrow G$ be the canonical morphism. Its image is also the image of $\varphi : G \times H \rightarrow G$, where*

$$\varphi(g, h) = \text{int}(g)h = ghg^{-1}.$$

The following conditions are equivalent:

- (i) H contains a Cartan subgroup C of G (XII 1).
- (i bis) H has the same reductive rank and the same nilpotent rank (XII 1) as G .
- (ii) H contains a maximal torus T of G , and $(\mathfrak{g}/\mathfrak{h})^T = 0$.
- (iii) The set of conjugates of H containing a given maximal torus is finite and nonempty, and H has finite index in its normalizer.
- (iv) There exists $a \in H(k)$ that belongs to only finitely many conjugates of H (or merely such that $\psi^{-1}(a)$ has an isolated point), and H has finite index in its normalizer.
- (iv bis) The morphism $\psi : X \rightarrow G$ is generically quasi-finite (that is, there is a dense open subset of X on which ψ is quasi-finite), and H has finite index in its normalizer.
- (v) There is a dense open subset U of G such that, for every $x \in U(k)$, the set of conjugates of H containing x is finite and nonempty; equivalently, $\psi : X \rightarrow G$ is dominant and generically quasi-finite.
- (vi) There is a dense open subset U of G such that every $x \in U(k)$ is contained in a conjugate of H ; equivalently, $\psi : X \rightarrow G$ is dominant.
- (vii) There exists $a \in H(k)$ such that the fixed subspace of $\mathfrak{g}/\mathfrak{h}$ under $\text{ad}_{\mathfrak{g}/\mathfrak{h}}(a)$ is zero.

Moreover, these conditions imply that H is its own connected normalizer: N is smooth and $\dim H = \dim N$. They also imply that $\psi : X \rightarrow G$ is generically étale.

Proof. By 2.0, $\dim X \leq \dim G$, with equality if and only if $\dim H = \dim N$, or equivalently if and only if H has finite index in N . The inequality $\dim X \leq \dim G$ shows that ψ is dominant if and only if it is dominant and generically quasi-finite, and also if and only if it is generically quasi-finite and $\dim X = \dim G$. Since the last equality is equivalent, by 2.0, to H having finite index in N , conditions (vi), (v), and (iv bis) are equivalent. The equivalence of (iv) and (iv bis) is immediate.

The equivalence of (i) and (i bis) follows directly from the definitions and is left to the reader. If H contains a Cartan subgroup C of G , it contains the maximal torus T of C , which is a maximal torus of G . Since C is the centralizer of T , its Lie algebra $\mathfrak{c}$ is

$$\mathfrak{c} = \mathfrak{g}^T$$

(II 5.2.3(ii)). Hence $\mathfrak{g}^T = \mathfrak{c} \subset \mathfrak{h}$, and I 4.7.3 gives the relation⁶

$$(x) \quad (\mathfrak{g}/\mathfrak{h})^T = 0.$$

⁶The French source has two evident typographical slips in this sentence, writing $\mathfrak{n}$ for $\mathfrak{c}$ and reversing the inclusion involving $\mathfrak{h}$ and $\mathfrak{g}^T$. The mathematically required relation is restored here.

Conversely, suppose that H contains the maximal torus T and that (x) holds, so that $\mathfrak{h} \supset \mathfrak{c}$. We claim that H contains the centralizer C of T , which will prove the equivalence of (i) and (ii). This follows from the next lemma.

Lemma 2.1.1. — *Let G be a smooth algebraic group over a field k , let T be a subgroup of multiplicative type of G , let C be its connected centralizer (equal to the centralizer of T when G is connected and T is a torus, by XII 6.6(b)), and let H be a smooth subgroup of G containing T . Then H contains C if and only if its Lie algebra $\mathfrak{h}$ contains the Lie algebra $\mathfrak{c}$ of C .*

Indeed, $\text{Centr}_G(T)$ is smooth over k by XI 2.4, and hence C is smooth over k . Likewise, $\text{Centr}_H(T)$ is smooth, and

$$\text{Centr}_H(T) = \text{Centr}_G(T) \cap H$$

has Lie algebra $\mathfrak{h} \cap \mathfrak{c}$. The hypothesis therefore implies that this smooth subgroup has the same Lie algebra as the smooth group $\text{Centr}_G(T)$, so it contains the connected component C of the latter. Thus H contains C . $\square$

We now prove the equivalence of (i) and (iii). If H contains the maximal torus T of G , we must show that $H \supset C$ (equivalently, that (x) holds) if and only if H has finite index in its normalizer and the set of conjugates of H containing T is finite. If $H \supset C$, then (x) implies, a fortiori,

$$(xx) \quad (\mathfrak{g}/\mathfrak{h})^H = 0.$$

The Lie algebra $\mathfrak{n}$ is the inverse image of the left-hand side under the canonical homomorphism $\mathfrak{g} \rightarrow \mathfrak{g}/\mathfrak{h}$ (II 5.2.3(i)). Consequently, by 2.0, (xx) says that $H = N^0$; in particular, H has finite index in its normalizer. Consider the diagram of subgroups

$$\begin{array}{ccccc} T & \longrightarrow & N(T) \cap H & \longrightarrow & H \\ & & \downarrow & & \downarrow \\ & & N(T) \cap N(H) & \longrightarrow & N(H) \\ & & \downarrow & & \\ & & N(T) & & \end{array}$$

By the conjugacy theorem for maximal tori in H (XII 6.6(a)), every conjugate of H containing T is obtained from H by conjugation by an element of $N(T)(k)$. Thus the set of such conjugates is in bijection with

$$(N(T)/(N(T) \cap N(H)))(k).$$

Since $H \supset C$, we have $N(T) \cap H \supset C$, so the preceding set is a quotient of $(N(T)/C)(k)$, which is finite. This proves (i) $\Rightarrow$ (iii).

Conversely, suppose (iii), so that $N(T)/(N(T) \cap N(H))$ and $N(H)/H$ are finite. The conjugacy theorem in H shows that the homomorphism

$$(N(T) \cap N(H))/(N(T) \cap H) \longrightarrow N(H)/H$$

induced by the preceding diagram is bijective on k -valued points (indeed, it is an isomorphism). Since the latter quotient is finite, so is the former. Hence $N(T) \cap H$ has finite

index in $N(T)$ and therefore contains $C = N(T)^0$. Thus $H \supset C$, and (i), (i bis), (ii), and (iii) are equivalent.

We next prove (ii) $\Rightarrow$ (vii). Both conditions are invariant under an extension $k \rightarrow k'$ with k' algebraically closed. We may therefore assume that k has infinite transcendence degree over its prime subfield. There is then an element $a \in T(k)$ such that the subgroup of $T(k)$ generated by a is Zariski dense in T . It follows that

$$(\mathfrak{g}/\mathfrak{h})^T = (\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)}.$$

The left-hand side is zero by hypothesis, which proves (vii).

The implication (vii) $\Rightarrow$ (vi) is contained in the following result, which sharpens 2.1.

Corollary 2.2. — *Let G be a smooth algebraic group over a field k , let H be a smooth algebraic subgroup, let N be its normalizer in G , and let*

$$\varphi : G \times H \longrightarrow G, \quad \varphi(g, h) = \text{ad}(g)h = ghg^{-1}.$$

Let $\psi : X = G \times^N H \rightarrow G$ be the morphism obtained from φ by passage to the quotient (cf. Section 1), and let $a \in H(k)$. The following conditions are equivalent:

- (i) φ is smooth at (e, a) .
- (ii) ψ is étale at $(\bar{e}, a)$, and N is smooth over k .
- (iii) $(\mathfrak{g}/\mathfrak{h})^{\text{ad}(a)} = 0$, where $\mathfrak{g}$ and $\mathfrak{h}$ are the Lie algebras of G and H .

These conditions imply $H^0 = N^0$.

The smoothness at a k -rational point of φ , a morphism between smooth k -preschemes, is equivalent to the surjectivity of the tangent map at that point. Under the usual identifications of the tangent spaces of G and H with their Lie algebras, a direct calculation gives

$$d\varphi(\xi, \eta) = (\text{id} - \text{ad}(a))\xi + \eta,$$

as a map from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is therefore equivalent to the surjectivity of $\text{id} - \text{ad}(a)$ on $\mathfrak{g}/\mathfrak{h}$, which is precisely (iii).

Condition (iii) implies, a fortiori, $(\mathfrak{g}/\mathfrak{h})^H = 0$, and hence $H^0 = N^0$ by 2.0 (the connectedness hypothesis made at the beginning of the section is not needed there). It follows that N is smooth and that $\dim H = \dim N$, whence $\dim X = \dim G$. Since $q : G \times H \rightarrow X$ is flat and $\psi \circ q = \varphi^7$ is smooth at (e, a) , the morphism ψ is smooth at $q(e, a) = (\bar{e}, a)$, and hence is étale there by dimension. We have proved (i) $\iff$ (iii) $\implies$ (ii). Conversely, (ii) implies (i), because the smoothness of N implies that of q . $\square$

In particular, under condition (vii) of 2.1 the morphism φ is smooth at (e, a) , so its image contains a nonempty open subset of G . Thus ψ is dominant, proving (vii) $\Rightarrow$ (vi).

It remains to prove (vi) $\Rightarrow$ (i). Suppose first that G is affine. Let U be a nonempty open subset of G such that every $x \in U(k)$ is contained in a conjugate of H , and let C be a Cartan subgroup of G . Applying the already proved implication (i) $\Rightarrow$ (iv) with C in place of H (Borel's "density theorem"), we find a conjugate of C meeting U . We may therefore assume that $U \cap C \neq \emptyset$; equivalently, there is a nonempty open subset V of C such that every $x \in V(k)$ belongs to a conjugate of H .

Write

$$C = T \cdot C_u,$$

⁷The source reverses this composition, printing $\psi = \varphi \circ q$. The displayed morphisms and the definition of φ require $\varphi = \psi \circ q$.

where T is the maximal torus of C , hence a maximal torus of G , and C_u is the unipotent part of C ; T lies in the center of C (*BIBLE* 6, Theorem 2). We may again assume that k has infinite transcendence degree over its prime subfield. Choose $t \in T(k)$ in the projection of V onto T , a nonempty open subset of T , so that $tC_u \cap V \neq \emptyset$, and choose it so that it generates a Zariski-dense subgroup of T . Every algebraic subgroup of G containing a product tu , with $t \in T(k)$ and $u \in C_u(k)$, contains both factors (*BIBLE* 4, Theorem 3). Taking $tu \in V(k)$, we conclude that a conjugate of H contains t , and therefore contains T . After replacing H by that conjugate, we may assume

$$T \subset H.$$

Let W be the inverse image in C_u of V under $u \mapsto tu$. For every $x \in (T \cdot W)(k)$, there is then a conjugate of H containing both T and x . As observed above, every such conjugate is of the form $\text{int}(g)H$ for some $g \in N(T)(k)$. Consider

$$f : N(T) \times H \longrightarrow G, \quad f(g, h) = \text{int}(g)h = ghg^{-1}.$$

The image of f contains TW . Since C has finite index in $N(T)$, the latter is a finite union of translates Cg_i , with $g_i \in N(T)(k)$. It follows that some dense open subset V' of $C = TC_u$ is contained in the image under f of $(Cg_i) \times H$. Replacing H by $\text{int}(g_i)H$, we may assume $g_i = e$, so that

$$f(C \times H) \supset V'.$$

Thus for every $u \in V'(k)$ there are $v \in C(k)$ and $h \in H(k)$ such that

$$v^{-1}hv = u, \quad \text{hence} \quad vuv^{-1} \in H(k).$$

Set

$$C' = C \cap H = \text{Centr}_H(T).$$

Since vuv^{-1} also belongs to C , it lies in C' , and hence $u \in \text{int}(v^{-1})C'$. Therefore the union of the conjugates of C' in C , by elements of $C(k)$, is dense. As in the preceding argument for the pair (G, H) , now applied to (C, C') , this implies that C' has finite index in its normalizer in C . By the lemma of *BIBLE* 7, Section 1, we obtain $C' = C$, and consequently $C \subset H$.⁸ This proves (vi) $\Rightarrow$ (i) when G is affine.

In the general case we argue by induction on $n = \dim G$, the assertion being trivial for $n = 0$. Let Z be the center of G . There are two cases.

- 1° If $\dim(Z \cap H) > 0$, set $G' = G/(Z \cap H)$. Then $\dim G' < n$, and condition (vi) for H implies the same condition for its image H' in G' . By induction, H' contains a Cartan subgroup C' of G' . Hence H contains the inverse image C of C' , which is a Cartan subgroup by XII 6.6(e).
- 2° If $\dim(Z \cap H) = 0$, the canonical morphism $H \rightarrow G/Z$ is finite. Since G/Z is affine by XII 6.1, H is affine. Every homomorphism from H to an abelian variety is therefore zero (indeed, every morphism of preschemes from H to an abelian variety is constant). This follows from the fact that a smooth connected affine algebraic group over an algebraically closed field is a rational variety, or simply from the fact that it is the union of its Borel subgroups (*BIBLE* 6, Theorem 5(b)); the structure theorems *BIBLE* 6.2 and 6.3 show readily that a smooth connected solvable affine group is a rational variety.

⁸The source prints $C = H$; only the inclusion $C \subset H$ follows and is used.

Apply Chevalley's structure theorem to G : the group G is an extension of an abelian variety A by a smooth affine group. Since H is affine, its image in A is zero. On the other hand, that image must be all of A , because the union of its conjugates in A is dense. Hence $A = 0$, so G is affine, reducing us to the case already proved.

This completes the proof of 2.1.

Corollary 2.3. — *Assume that the equivalent conditions of 2.1 hold.*

- a) Let $k(X)$ and $k(G)$ be the fields of rational functions of X and G , respectively. Then $k(X)$ is a finite separable extension of $k(G)$. Denote its degree by d .
- b) Let T be a maximal torus of G contained in H , which exists by 2.1(ii), and let C be the corresponding Cartan subgroup of G . Then $C \subset H$. Moreover, $\text{Norm}_G(T)$ is a smooth subgroup of G , and

$$\text{Norm}_G(T) \cap \text{Norm}_G(H) = \text{Norm}_{\text{Norm}_G(H)}(T)$$

is a smooth subgroup of it, of finite index equal to the degree d defined in (a). The number of conjugates of H containing a given maximal torus, or a given Cartan subgroup, is d .

- c) Let U be the largest open subset of G such that $\psi : X \rightarrow G$ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U.$$

Then U is dense. For $g \in G(k)$, the following are equivalent:

- (a) $g \in U(k)$;
- (b) exactly d conjugates of H contain g ;
- (c) there are at least d distinct conjugates H_i of H containing g , and for every i ,

$$(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0, \quad \mathfrak{h}_i = \text{Lie}(H_i).$$

Assertion (a) follows from the generic étaleness of ψ , stated at the end of 2.1. This also shows that the open set U in (c) is nonempty, hence dense, and gives the two stated characterizations of its k -points: here one uses that ψ is separated, that X is integral, that G is integral and normal (SGA 1, I 10.11), and that $(\mathfrak{g}/\mathfrak{h}_i)^{\text{ad}(g)} = 0$ means that ψ is étale at the point $x_i \in \psi^{-1}(g)$ corresponding to H_i .

If H contains the maximal torus T of G , the centralizers of T in H and G have the same dimension and are smooth and connected (XII 6.6(b)); they are therefore equal, proving that $C \subset H$.⁹ The normalizer of T in any smooth group containing it is smooth (XI 2.4 bis), so $\text{Norm}_G(T)$ and $\text{Norm}_{\text{Norm}_G(H)}(T)$ are smooth. Recall also that $N = \text{Norm}_G(H)$ is smooth by the last assertion of 2.1. Furthermore, $\text{Norm}_N(T)$ contains C , which has finite index in $\text{Norm}_G(T)$; hence $\text{Norm}_N(T)$ itself has finite index in $\text{Norm}_G(T)$.

The conjugacy theorem for maximal tori in H shows that this index is the number of conjugates of H containing T , equivalently the number containing C . The union of the conjugates of C in G is dense, by 2.1(i) $\Rightarrow$ (vi) applied to C in place of H , and U is stable under inner automorphisms. Hence $C \cap U \neq \emptyset$. Arguing as in the proof of 2.1(vi) $\Rightarrow$ (ii), and making a harmless extension of the base field if necessary, one finds $g \in (C \cap U)(k)$ such that every conjugate of H containing g contains T , and hence C . Thus the conjugates

⁹The source prints the tautological $C \subset G$; equality of the two centralizers gives $C \subset H$.

of H containing C are precisely those containing g ; since $g \in U(k)$, their number is d . This proves (b).

In fact, the argument shows that the set of conjugates of H containing T is a homogeneous set under the group of rational points of

$$W_G(T) = \text{Norm}_G(T) / \text{Centr}_G(T).$$

In particular,

$$d \leq \text{the order of the Weyl group of } G.$$

Corollary 2.4. — *With the notation of 2.1, the following conditions are equivalent:*

- (i) $\psi : X \rightarrow G$ is birational.
- (ii) Exactly one conjugate of H contains a given Cartan subgroup of G .
- (iii) H contains a Cartan subgroup C of G , and $\text{Norm}_G(H) \supset \text{Norm}_G(C)$.
- (iv) There is a nonempty open subset V of G such that every $g \in V(k)$ belongs to exactly one conjugate of H .

This follows immediately from 2.1 and 2.3.

Corollary 2.5. — *Assume that the conditions of 2.4 hold, and let $g \in G(k)$. The following conditions are equivalent:*

- (i) $g \in U(k)$, where U is defined in 2.3(c); that is, g is contained in exactly one conjugate of H .
- (ii) The set of conjugates of H containing g is finite and nonempty.
- (iii) The scheme $\psi^{-1}(g)$, the “scheme of conjugates of H containing g ,” has an isolated point.
- (iv) There is a conjugate H' of H containing g such that

$$(\mathfrak{g}/\mathfrak{h}')^{\text{ad}(g)} = 0, \quad \mathfrak{h}' = \text{Lie}(H').$$

Finally, U is also the largest open subset of G on which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

The equivalence of (i), (ii), and (iii), together with the last assertion, follows from the “Main Theorem”¹⁰ applied to the birational morphism $\psi : X \rightarrow G$, since G is normal. Their equivalence with (iv) follows immediately from Corollary 2.2, which characterizes the points of X at which ψ is étale.¹¹

Theorem 2.6. — *Let G be a smooth connected algebraic group over an algebraically closed field k , let C be a Cartan subgroup associated with a maximal torus T , and put*

$$N = \text{Norm}_G(C) = \text{Norm}_G(T) \quad (\text{cf. XII 8.4}).$$

Let $X = G \times^N C$, where N acts on the left factor G by right translations and on the right factor C by inner automorphisms, and let $\psi : X \rightarrow G$ be the canonical morphism.

¹⁰Editor’s note in the French source: Zariski’s Main Theorem.

¹¹The source cites the final assertion of 2.1 here; the pointwise étaleness criterion is Corollary 2.2.

- a) ψ is birational.
 b) Let U be the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U$$

(cf. 2.5), and let

$$\rho = \rho_\nu(G) = \dim C$$

be the nilpotent rank of G . For every $g \in G(k)$, the multiplicity of the eigenvalue 1 in the action of $\text{ad}(g)$ on $\mathfrak{g}$ is at least ρ ; it equals ρ if and only if $g \in U(k)$.

Proof. Condition 2.1(i) holds, so 2.4(iii) $\Rightarrow$ (i) proves (a). In *BIBLE* 7, when G is affine, the points of $U(k)$ are called the *regular points* of $G(k)$. We shall use the same terminology and call U the *open subset of regular points* of G . Notice that the proof in *BIBLE* that this set is itself open is incorrect; the present Exposé establishes its openness under more general hypotheses.

To prove (b), for each $g \in G(k)$ introduce the characteristic polynomial

$$P(\text{ad}(g), t) = t^n + c_1(g)t^{n-1} + \cdots + c_n(g).$$

Replacing k by an arbitrary k -algebra shows immediately that the functions $c_i(g)$ arise from uniquely determined sections

$$c_i \in \Gamma(G, \mathcal{O}_G).$$

Suppose that $g \in G(k)$ belongs to a Cartan subgroup, which we may take to be C ; this includes the case of a regular element. By 2.5(iv),

$$(\mathfrak{g}/\mathfrak{c})^{\text{ad}(g)} = 0$$

if and only if g is regular, where $\mathfrak{c} = \text{Lie}(C)$. Since C is nilpotent, $\text{ad}_{\mathfrak{c}}(g)$ has only the eigenvalue 1. It follows that the multiplicity of the eigenvalue 1 in $\text{ad}_{\mathfrak{g}}(g)$ is at least ρ , and is exactly $\dim C = \rho$ if and only if g is regular.

In particular, the characteristic polynomial above is divisible by $(t-1)^\rho$. Divisibility by $(t-1)^\rho$ is expressed by linear relations, with integer coefficients, among the coefficients of the polynomial. These relations hold for $g \in U(k)$, where U is dense; since G is reduced, they therefore hold everywhere. Thus¹²

$$(\dagger) \quad t^n + c_1 t^{n-1} + \cdots + c_n = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho})$$

in the polynomial ring over $\Gamma(G, \mathcal{O}_G)$. In particular, for every field extension K/k and every $g \in G(K)$, the eigenvalue 1 of $\text{ad}(g)$ has multiplicity at least ρ .

We have already seen that equality holds when g is regular; let us prove the converse. Suppose first that G is affine, and write

$$g = g_s g_u$$

as the product of its semisimple and unipotent parts (*BIBLE* 4, Section 4). Then

$$\text{ad}(g) = \text{ad}(g_s) \text{ad}(g_u)$$

¹²The French source prints c_0 in place of the omitted tail $+\cdots + c_n$ on the left-hand side of $(\dagger)$. The characteristic polynomial displayed immediately above determines the correction made here.

is the corresponding decomposition of $\mathrm{ad}(g)$ (loc. cit., corollary to Theorem 3). Hence $\mathrm{ad}(g)$ and $\mathrm{ad}(g_s)$ have the same eigenvalues, counted with multiplicity; in particular, the eigenvalue 1 has the same multiplicity in both.

By *BIBLE* 7, Theorem 2, Corollary 1, the element g is regular if and only if g_s is regular. We may therefore assume that $g = g_s$ is semisimple. It is then contained in a maximal torus by *BIBLE* 6, Theorem 5(c), and hence in a Cartan subgroup, a case already treated. This proves (b) when G is affine.

In general, let $Z = \mathrm{Centr}(G)_{\mathrm{red}}$. By XII 6.6(e), the Cartan subgroups of G are the inverse images of those of $G' = G/Z$. Thus g is regular in G if and only if its image g' is regular in G' . Since Z is smooth, the Lie algebra of G' is $\mathfrak{g}' = \mathfrak{g}/\mathfrak{z}$, where $\mathfrak{z} = \mathrm{Lie}(Z)$, and $\mathrm{ad}(g')$ is the endomorphism induced by $\mathrm{ad}(g)$ on $\mathfrak{g}/\mathfrak{z}$. Consequently, the multiplicity of the eigenvalue 1 in $\mathrm{ad}(g)$ is

$$d = \dim Z$$

plus its multiplicity in $\mathrm{ad}(g')$. The former therefore equals the nilpotent rank of G if and only if the latter equals the nilpotent rank of G' . We are reduced to G' , which is affine by XII 6.1, and that case has already been proved. This completes the proof of 2.6.

Corollary 2.7. — *With the notation of the preceding proof,¹³ let*

$$b = 1 + b_1 + \cdots + b_{n-\rho} \in \Gamma(G, \mathcal{O}_G).$$

Then the open subset of regular points of G is

$$U = G_b,$$

the set of points of G at which b is invertible. In particular, U is an affine open subset when G is affine.

Corollary 2.8. — *Let H be a smooth connected algebraic subgroup of G containing a Cartan subgroup of G .*

- a) Let C be an algebraic subgroup of H . Then C is a Cartan subgroup of H if and only if it is a Cartan subgroup of G .
- b) Let $g \in G(k)$, and let d be the integer introduced in 2.3. Then g is a regular point of G if and only if there are exactly d conjugates H_i of H containing g , and g is a regular element of every H_i . Equivalently, there are at most d conjugates of H containing g , and g is regular in one of them. When these conditions hold, if C is the unique Cartan subgroup of G containing g , then the conjugates of H containing g are precisely the conjugates of H containing C .
- c) Let $g \in H(k)$. Then g is regular in G if and only if it is regular in H and

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(g)} = 0.$$

Proof. a) Under either hypothesis on C , its unique maximal torus T is a maximal torus of both H and G , since H has the same reductive rank as G . The smooth connected groups $\mathrm{Centr}_H(T) \subset \mathrm{Centr}_G(T)$ have the same dimension, and therefore are equal. Hence C is equal to one of these two centralizers if and only if it is equal to the other, proving (a).

¹³Editor's note in the French source: cf. (†) above.

b) Suppose first that g is regular in G , and let C be the unique Cartan subgroup of G containing g . By 2.3(b), exactly d conjugates H_i of H contain C . Since

$$(\mathfrak{g}/\mathfrak{c})^{\mathrm{ad}(g)} = 0,$$

the endomorphism $\mathrm{ad}(g)$ has no eigenvalue 1 on $\mathfrak{g}/\mathfrak{c}$, and a fortiori

$$(\mathfrak{g}/\mathfrak{h}_i)^{\mathrm{ad}(g)} = 0.$$

By 2.3(c), exactly d conjugates of H contain g , namely the H_i . A Cartan subgroup of such an H_i containing g is a Cartan subgroup of G by (a), and hence is C . Thus g is regular in every H_i .

Conversely, suppose that at most d conjugates H_i of H contain g , and that g is regular in one of them, which we may take to be H . Since g is regular in H , it belongs to a unique Cartan subgroup C of H ; by (a), C is a Cartan subgroup of G . Let C' be any Cartan subgroup of G containing g . By 2.3(b), exactly d conjugates of H contain C' . They all contain g , so they are precisely the H_i ; in particular, H contains C' . Thus C and C' are two Cartan subgroups of H containing the same regular element g of H , and therefore $C = C'$. Hence g is regular in G .¹⁴ $\square$

c) For an endomorphism u of a finite-dimensional vector space, let $\nu(u)$ denote the nullity¹⁵ of $\mathrm{id} - u$. Then

$$\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \nu(\mathrm{ad}(g)_{\mathfrak{h}}) + \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}).$$

The two terms on the right are respectively at least the nilpotent rank of H , which equals the nilpotent rank ρ of G by (a), and at least 0. Hence $\nu(\mathrm{ad}(g)_{\mathfrak{g}}) = \rho$ if and only if

$$\nu(\mathrm{ad}(g)_{\mathfrak{h}}) = \rho, \quad \nu(\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}) = 0.$$

This says precisely that g is regular in G if and only if it is regular in H and $\mathrm{ad}(g)_{\mathfrak{g}/\mathfrak{h}}$ has no nonzero invariants. $\square$

Remarks 2.9. — In the statement of 2.1, condition (iii) cannot be weakened by assuming only that H contains a maximal torus and has finite index in its normalizer, even if one also requires that normalizer to be smooth, that is, that $H = N^0$, and even when G is affine and solvable. An example is the group G of matrices of the form

$$g = \begin{pmatrix} t & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix},$$

and the subgroup H formed by the matrices above for which $b = c = 0$. The Cartan subgroup of G here consists of the matrices g for which $a = c = 0$.

Remarks 2.10. — Let G be a smooth algebraic group over k , and let H be a smooth algebraic subgroup; we no longer assume that G and H are connected. Suppose that H^0 contains a Cartan subgroup of G^0 . Then

$$(\mathfrak{g}/\mathfrak{h})^H \subset (\mathfrak{g}/\mathfrak{h})^{H^0} = 0,$$

so $H^0 = N^0$, where N is the normalizer of H ; in particular, N is smooth.

¹⁴The source prints “regular in C ” at this conclusion; the argument proves regularity in G .

¹⁵Editor’s note in the French source: that is, the dimension of the nullspace of $\mathrm{id} - u$.

Nevertheless, even with G connected, examples are easily constructed in which H has a connected component H_i such that

$$(\mathfrak{g}/\mathfrak{h})^{\mathrm{ad}(h)} \neq 0 \quad \text{for every } h \in H_i(k).$$

Equivalently, the morphism

$$G \times H_i \longrightarrow G, \quad (g, h) \longmapsto \mathrm{ad}(g) \cdot h$$

is nowhere étale, or even quasi-finite. One may take, for example, H to be the normalizer of a maximal torus in $\mathrm{SL}(2)_k$.

Similarly, even if the image H'' of H in the finite group $G'' = G/G^0$ is all of G'' , the union of the conjugates of H in G need not be dense. For example, take G to be the semidirect product of $\mathbb{Z}/2\mathbb{Z}$ with $G^0 = \mathrm{SL}(2)_k$, on which it acts by “symmetry,” and take H to be the semidirect product $\mathbb{Z}/2\mathbb{Z} \cdot T$, where T is a maximal torus of G^0 .

On the other hand, if one does not assume in advance that H^0 contains a Cartan subgroup of G^0 , but assumes that the union of the conjugates of H in G is dense, then H^0 necessarily contains a Cartan subgroup of G^0 . To see this, one may plainly assume that G is connected and repeat the proof of 2.1(vi) $\Rightarrow$ (i), which does not require H to be connected.

3 The case of an arbitrary base prescheme

Suppose first that the base is a field k , not necessarily algebraically closed. Since conditions 2.1 (i bis), (iv bis), (v), and (vi) are invariant under extension of the base field, passage to the algebraic closure $\bar{k}$ of k shows that they are equivalent to one another, and equivalent to the assertion that $H_{\bar{k}}$ contains a Cartan subgroup of $G_{\bar{k}}$. When this condition is satisfied, it is still true (with the notation of 2.3) that $k(X)$ is a finite separable extension of $k(G)$, of degree d independent of every extension of the base. If U is the largest open subset of G such that ψ induces a finite étale morphism

$$\psi^{-1}(U) \longrightarrow U,$$

then the formation of U commutes with extension of the base field. If ψ is birational, then U is also the largest open subset of G such that ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U;$$

and if $g \in U(k)$, there is one and only one subgroup H' of G , conjugate to H over $\bar{k}$, such that $H' \ni g$.

A point $g \in G(k)$ is called *regular* if it is regular as an element of

$$G(\bar{k}) = G_{\bar{k}}(\bar{k}).$$

More generally, the construction of 2.7 gives an open subset of G , whose formation commutes with every extension of the base field, called the *open subset of regular points* of G . It is also characterized by the following property: for every algebraically closed extension K of k and every $g \in G(K)$, the point g is regular in G_K if and only if $g \in U(K)$. If $g \in U(k)$ ¹⁶, then g is contained in one and only one Cartan subgroup of G , as we shall prove below under more general hypotheses.

¹⁶Editor’s note: $G(k)$ has been corrected to $U(k)$.

Let G be a smooth, separated group prescheme of finite type, with connected fibers over the prescheme S , and consider the functor

$$\mathcal{C} : (\text{Sch})/S^\circ \longrightarrow (\text{Ens})$$

defined by

$$\mathcal{C}(S') = \left\{ \text{Cartan subgroups (XII 3.1) of } G_{S'} \right\}.$$

Suppose that this functor is representable by a prescheme smooth over S . In Exposé XV we give a necessary and sufficient condition for this to hold, but we already know that the hypothesis is satisfied if G is affine over S with locally constant reductive rank (XII 3.3), or, more generally, if G locally admits a maximal torus for the fpqc topology (XII 7.1 a)); this is the case, for example, if S is the spectrum of a field.

Let X be the Cartan subgroup of the $\mathcal{C}$ -group prescheme $G_{\mathcal{C}}$, the “universal Cartan subgroup” of G . Thus, as a prescheme over S , X represents the functor

$$X(S') = \left\{ (C, g) \mid \begin{array}{l} C \text{ is a Cartan subgroup of } G_{S'}, \\ g \text{ is a section of } C \text{ over } S' \end{array} \right\}.$$

Consider the canonical projection morphism $(C, g) \mapsto g$,

$$\psi : X \longrightarrow G.$$

We then have the following theorem.

Theorem 3.1. *Under the preceding hypotheses on G , and with the preceding notation, let U be the set of $g \in G$ such that g is a regular element of its fiber G_s . Then U is open, and it is also the largest open subset U of G such that ψ induces an isomorphism*

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Let us first prove that U is open. By the representability of $\mathcal{C}$ as a prescheme smooth over S , and since its structure morphism is plainly surjective, it follows at once that G locally admits a Cartan subgroup for the étale topology and that the nilpotent rank of the fibers of G is locally constant. The same holds for the dimension of the fibers of G . After localizing on S , we may therefore suppose that these two integers are constant, say ρ and n . Consider the Killing polynomial

$$P_G(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t], \quad A = \Gamma(G, \mathcal{O}_G).$$

The restriction of this polynomial to the fibers G_s of G , and in particular to the fibers at the maximal points of S , is divisible by $(t-1)^\rho$. This says that certain linear combinations of the c_i , with integer coefficients, vanish on the fibers G_s . When S is reduced, as we may assume in order to prove that U is open, it follows that these combinations themselves vanish. Hence the Killing polynomial itself is divisible by $(t-1)^\rho$, say¹⁷

$$P_G(t) = (t-1)^\rho (t^{n-\rho} + b_1 t^{n-\rho-1} + \cdots + b_{n-\rho}).$$

Let b be the sum of the coefficients $b_0 = 1, b_1, \dots, b_{n-\rho}$ of the second factor. By 2.7, applied to the fibers of G , we have

$$U = G_b,$$

¹⁷The source prints the undefined exponent $n - r - 1$. It is corrected to $n - \rho - 1$, as required by the degree $n - \rho$ of the second factor.

which proves that U is open.

To prove that $\psi^{-1}(U) \rightarrow U$ is an isomorphism, SGA 1, I 5.7 reduces us to checking this fiber by fiber. We are thereby reduced to the case of a base field, which may be assumed algebraically closed. There is then a Cartan subgroup C of G . If N is its normalizer, the conjugacy theorem XII 7.1 a) and b) identifies $\mathcal{C}$ with G/N , and the morphism $\psi : X \rightarrow G$ considered here is precisely the one defined in Section 2. The conclusion follows from 2.6 b). The same argument also shows that U is the largest open subset of G for which ψ induces an isomorphism $\psi^{-1}(U) \rightarrow U$.

Corollary 3.2. *Under the hypotheses of 3.1, let g be a regular section of G , that is, suppose that $g(s)$ is a regular point of G_s for every $s \in S$. Then there is one and only one Cartan subgroup C of G such that g is a section of C .*

Indeed, the hypothesis on g says that g is a section of U , while the conclusion says that it admits a unique lift to a section of X . This is simply another way of saying that $\psi^{-1}(U) \rightarrow U$ is an isomorphism.

Observe now that the open subset $\psi^{-1}(U)$ of the Cartan subgroup X of $G_{\mathcal{C}}$ is precisely the open subset of X consisting of the points regular in $G_{\mathcal{C}}$, by which we mean regular in their fibers. We thus obtain a natural “fibration” of the dense open subset U of regular points of G over the prescheme $\mathcal{C}$; its fibers are dense open subsets of Cartan subgroups of the fibers of $G_{\mathcal{C}}$, namely their open subsets of points regular in G . For example, we obtain the following result, which will be considerably sharpened in the next Exposé.

Corollary 3.3. *Let G be a smooth connected algebraic group over the field k , and let $\mathcal{T}$ be the scheme of maximal tori of G , which is isomorphic to the scheme of Cartan subgroups of G . Then the function field $k(G)$ of G is isomorphic to the function field of a smooth connected affine nilpotent algebraic group C over the function field $k(\mathcal{T})$ of $\mathcal{T}$, namely*

$$C = \text{“the generic Cartan subgroup of } G \text{”}.$$

If G is affine and has unipotent rank zero, that is, if the Cartan subgroups of $G_{\bar{k}}$ ¹⁸ are tori, then $k(G)$ is a unirational extension of $k(\mathcal{T})$.

Of course, by the generic Cartan subgroup of G we mean, by an abuse of language, the Cartan subgroup of $G_{k(\mathcal{T})}$ that is the generic fiber of X over $\mathcal{T}$. It remains only to prove the last assertion of 3.3, which is contained in the following well-known result, due to Chevalley.

Lemma 3.4. *Let k be a field, T a torus over k , and $k(T)$ the field of rational functions on T . Then $k(T)$ is a unirational extension of k , that is, it is contained in a purely transcendental extension of k .*

Indeed, let k' be a finite separable extension of k that splits¹⁹ T (X 1.4). Then $T \otimes_k k'$ is a rational variety: it contains a dense open subset isomorphic to a dense open subset of affine n -space $\mathbb{A}_{k'}^n$. Hence

$$T' = \coprod_{\text{Spec}(k')/\text{Spec}(k)} T_{k'}/\text{Spec}(k')$$

is a rational variety, since it has a dense open subset isomorphic to a dense open subset of

$$\coprod_{\text{Spec}(k')/\text{Spec}(k)} \mathbb{A}_{k'}^n,$$

¹⁸The source prints the undefined base field $\bar{K}$; it is corrected to $\bar{k}$.

¹⁹Editor’s note: “splits” (?), or: “over which T is split”.

which is isomorphic to affine mn -space over k , where $m = [k' : k]$. Consider the norm homomorphism from T' to T , which is defined whenever T is a commutative group scheme over k . The composite

$$T \longrightarrow T' \longrightarrow T$$

is the m -th power map on T , and is therefore dominant. Consequently $T' \rightarrow T$ is dominant, proving that T is unirational.

Return to the hypotheses of 3.1, but suppose in addition that G locally admits a maximal torus for the fpqc topology (XII 7.1). Let Y be the maximal torus of the Cartan subgroup X of $G_{\mathcal{C}}$. The morphism $\psi : X \rightarrow G$ then induces a morphism $Y \rightarrow G$ whose set-theoretic image consists of the semisimple elements of the fibers of G (XII 8). Finally, 3.1 shows that the restriction of ψ to the open subset Y^{reg} of regular points of Y induces a closed immersion

$$Y^{\text{reg}} \longrightarrow U = G^{\text{reg}}.$$

Making explicit the functorial meaning of $Z = Y^{\text{reg}}$, we obtain the following.

Corollary 3.5. *Let G be a smooth, separated S -group prescheme of finite type, with connected fibers over the prescheme S , and suppose that G locally admits a maximal torus for the fpqc topology. Let*

$$Z : (\text{Sch})_S^{\circ} \longrightarrow (\text{Ens})$$

be the functor defined by

$$Z(S') = \left\{ \begin{array}{l} \text{regular sections of } G_{S'} \text{ over } S' \text{ that are} \\ \text{contained in a maximal torus of } G_{S'} \end{array} \right\}.$$

Then Z is representable by a closed subprescheme, smooth over S , of the open subset $U = G^{\text{reg}}$ of G introduced in 3.1.

To conclude, let us record the following result, which sharpens the density theorem 2.1 (i) $\Rightarrow$ (vi).

Corollary 3.6. *Under the hypotheses of 3.5, let C be a Cartan subgroup of G , and consider the morphism*

$$\varphi : Z \times C \longrightarrow G$$

defined by

$$\varphi(g, h) = \text{ad}(g)h = ghg^{-1}.$$

Then φ is dominant.

It plainly suffices to prove the assertion fiber by fiber, which reduces us to the case where S is the spectrum of an algebraically closed field k . Let T be the maximal torus of C , let $t_0 \in T(k)$ be regular in G , and let $c_0 \in C(k)$ be regular in G . Consider $\varphi^{-1}(\varphi(t_0, c_0))$. Its k -rational points are the pairs (t, c) , with $t \in Z(k)$ and $c \in C(k)$, such that

$$\text{ad}(t)c = \text{ad}(t_0)c_0, \quad \text{that is,} \quad c = \text{ad}(t^{-1}t_0)c_0.$$

They are therefore in bijection with the $t \in Z(k)$ such that $\text{ad}(t^{-1}t_0)c_0 \in C$, or equivalently, since c_0 is regular, such that $t^{-1}t_0 \in N$, where N is the normalizer of C ; in other words, $t \in N$. Restricting to those $t \in Z(k)$ that belong to $C(k)$ gives an open and closed part of this fiber. We have therefore found a connected component of $\varphi^{-1}(\varphi(t_0, c_0))$ isomorphic to T . The preceding set-theoretic argument indeed gives an isomorphism of schemes: one simply replaces k -valued points by points with values in an arbitrary k -prescheme.

It follows that the generic fiber of φ has dimension at most $\dim T$, and therefore

$$\dim \operatorname{Im}(\varphi) \geq \dim(Z \times C) - \dim T = \dim Z + \dim C - \dim T.$$

But

$$\dim Z = \dim Y = \dim C + \dim T = \dim G - \dim C + \dim T.$$

Thus $\dim \operatorname{Im}(\varphi) \geq \dim G$, and φ is dominant.

Q.E.D.

Remarks 3.7. The argument shows in addition that the connected component at (t_0, c_0) of the fiber $\varphi^{-1}(\varphi(t_0, c_0))$ is isomorphic to T . In particular, it is smooth over k and has the same dimension as the generic fiber. It follows that φ is in fact smooth at (t_0, c_0) , something that one should also be able to verify by calculating the tangent map. Consequently, under the hypotheses of 3.6, the induced morphism

$$Z \times_S C^{\operatorname{reg}} \longrightarrow G^{\operatorname{reg}}, \quad C^{\operatorname{reg}} = C \cap G^{\operatorname{reg}},$$

is smooth. Likewise, the analogous morphism

$$Z \times T^{\operatorname{reg}} \longrightarrow Z,$$

where T is a maximal torus of G , is smooth. More generally, for every smooth connected normal algebraic subgroup H of C containing an element $c_0 \in G(k)$ that is regular in G , the image of $Z \times H \rightarrow G$ is dense in the image of $G \times H \rightarrow G$.

4 Lie algebras over a field: rank, regular elements, Cartan subalgebras

In the remainder of this Exposé we take up the theory developed by Chevalley in his book *Théorie des Groupes de Lie III* (Act. Sc. Ind. 1226, Paris 1955); the methods of schemes allow us to dispense with the hypothesis of characteristic zero. We begin by recalling in this section several well-known notions and results.

Let $\mathfrak{g}$ be a Lie algebra over a ring k . For every $a \in \mathfrak{g}$, denote by $\operatorname{ad}(a)$ the endomorphism

$$\operatorname{ad}(a) \cdot x = [a, x]$$

of $\mathfrak{g}$; it is a derivation of the Lie algebra $\mathfrak{g}$. For every derivation D of $\mathfrak{g}$, the nil-space of D , that is, the union of the kernels of the iterates of D , is a Lie subalgebra of $\mathfrak{g}$, as follows from the Leibniz formula

$$D^m([x, y]) = \sum_{0 \leq p \leq m} \binom{m}{p} [D^p x, D^{m-p} y].$$

We set

$$\operatorname{Nil}(a, \mathfrak{g}) = \text{nil-space of } \operatorname{ad}(a) = \bigcup_{m \geq 0} \operatorname{Ker} \operatorname{ad}(a)^m.$$

When no confusion can arise, we write simply $\operatorname{Nil}(a)$, and call it the *nil-space of a in $\mathfrak{g}$* .

Proposition 4.1. *For every $a \in \mathfrak{g}$, its nil-space $\operatorname{Nil}(a, \mathfrak{g})$ is a Lie subalgebra of $\mathfrak{g}$, equal to its own normalizer.*

It remains only to prove that it is its own normalizer. In other words, every element of $\mathfrak{g}/\text{Nil}(a, \mathfrak{g})$ annihilated by the adjoint representation of $\text{Nil}(a, \mathfrak{g})$ on this quotient must be zero. This is immediate, since every element of the quotient annihilated by $\text{ad}(a)^{20}$ is zero.

For the remainder of this section, suppose that k is a field and that $\mathfrak{g}$ is finite-dimensional over k . We denote by $W(\mathfrak{g})$ the scheme over k defined by $\mathfrak{g}$; its points in a k -algebra A are the elements of $\mathfrak{g} \otimes_k A$. If $a \in \mathfrak{g}$, the characteristic polynomial of $\text{ad}(a)$ is also called the *characteristic polynomial*, or the *Killing polynomial*, of a in $\mathfrak{g}$:

$$P_{\mathfrak{g}}(a, t) = t^n + c_1(a)t^{n-1} + \cdots + c_n(a),$$

where $n = \text{rank}_k \mathfrak{g}$ and $c_i(a) \in k$. Taking the same polynomial for $a \in \mathfrak{g} \otimes_k A^{21}$, where A is any k -algebra, we see that the $c_i(a)$ arise from uniquely determined sections c_i of the structure sheaf of $W(\mathfrak{g})$, that is, from elements of the symmetric algebra

$$A = \text{Sym}_k(\mathfrak{g}^\vee),$$

where $\mathfrak{g}^\vee$ is the dual of the k -module $\mathfrak{g}$. When k is infinite, the c_i are determined by the corresponding polynomial functions $\mathfrak{g} \rightarrow k$, but this is no longer true when k is finite.

Let r be the largest integer such that the Killing polynomial

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_n \in A[t]$$

is divisible by t^r ; thus

$$P_{\mathfrak{g}}(t) = t^n + c_1 t^{n-1} + \cdots + c_{n-r} t^r, \quad c_{n-r} \neq 0.$$

The integer r is called the *nilpotent rank* of the Lie algebra $\mathfrak{g}$. It is invariant under extension of the base field.

Proposition 4.2. *Let r be the nilpotent rank of $\mathfrak{g}$, and let $a \in \mathfrak{g}$. Then*

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r,$$

with equality if and only if

$$c_{n-r}(a) \neq 0.$$

In this case $\text{Nil}(a, \mathfrak{g})$ is a nilpotent Lie algebra. We shall see the converse in 5.7 b) when $\mathfrak{g}$ is the Lie algebra of an algebraic group G smooth over k .

The first assertion is immediate, since by definition

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) = \text{the multiplicity of the zero root in } P_{\mathfrak{g}}(a, t).$$

Suppose now that $c_{n-r}(a) \neq 0$. We prove that $\text{Nil}(a)$ is nilpotent, which is equivalent to saying that $\text{ad}_{\text{Nil}(a)}(x)$ is a nilpotent endomorphism for every $x \in \text{Nil}(a)$. We may suppose that k is algebraically closed. Since

$$\text{ad}(a)_{\mathfrak{g}/\text{Nil}(a)}$$

is injective, there is a nonempty open subset $U \subset W(\text{Nil}(a))$ such that, for every $x \in U(k)$, the endomorphism $\text{ad}(x)_{\mathfrak{g}/\text{Nil}(a)}$ is injective and hence $\text{Nil}(x) \subset \text{Nil}(a)$. We may further

²⁰The source prints $\text{Ad}(a)$; because $a \in \mathfrak{g}$, the endomorphism here is $\text{ad}(a)$.

²¹Editor's note: $W \otimes_k A$ has been corrected to $\mathfrak{g} \otimes_k A$.

suppose that U is contained in the open subset on which c_{n-r} does not vanish; that open subset is nonempty because $c_{n-r}(a) \neq 0$. Then $\text{Nil}(x)$ and $\text{Nil}(a)$ have the same dimension, so

$$\text{Nil}(x) = \text{Nil}(a).$$

It follows that $\text{ad}(x)_{\text{Nil}(a)}$ is nilpotent for every $x \in U(k)$. By the principle of extension of algebraic identities, this remains true for every $x \in \text{Nil}(a)$, and hence $\text{Nil}(a)$ is nilpotent.

An element $a \in \mathfrak{g}$ is called *regular* if $c_{n-r}(a) \neq 0$, that is, if

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) = r.$$

When k is infinite, this also says that $\text{rank}_k \text{Nil}(a, \mathfrak{g})$ is as small as possible as a varies in $\mathfrak{g}$. In every case, the notion of a regular element of $\mathfrak{g}$ is invariant under extension of the base field. The set of regular points of $W(\mathfrak{g})$, meaning those that arise from regular points with values in a suitable extension of k , is open, since it is

$$W(\mathfrak{g})_{c_{n-r}},$$

the set of points at which c_{n-r} is invertible.

Corollary 4.3. *Let a be a regular element of $\mathfrak{g}$, and let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$ containing a . Then $\mathfrak{h}$ is nilpotent if and only if*

$$\mathfrak{h} \subset \text{Nil}(a, \mathfrak{g}).$$

In particular, $\text{Nil}(a, \mathfrak{g})$ is a maximal nilpotent subalgebra of $\mathfrak{g}$.

Since $\text{Nil}(a)$ is nilpotent, the inclusion $\mathfrak{h} \subset \text{Nil}(a)$ implies that $\mathfrak{h}$ is nilpotent. Conversely, if $\mathfrak{h}$ is nilpotent, it is contained in the nil-space of its element a ; that is, $\mathfrak{h} \subset \text{Nil}(a)$.

Proposition 4.4. *Suppose that k is infinite. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{g}$. Consider the following conditions:*

- (i) $\mathfrak{d}$ is maximal nilpotent and contains a regular element of $\mathfrak{g}$.
- (i bis) $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ for some regular element a of $\mathfrak{g}$.
- (ii) $\mathfrak{d}$ is nilpotent and $\mathfrak{d} = \text{Nil}(a, \mathfrak{g})$ for some $a \in \mathfrak{g}$.
- (ii bis) $\mathfrak{d}$ is nilpotent, and there exists $a \in \mathfrak{d}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.
- (iii) $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Then

$$(i) \iff (i \text{ bis}) \implies (ii) \iff (ii \text{ bis}) \iff (iii).$$

We shall see in 5.7 a) that if $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group, then all the preceding conditions are equivalent.

The equivalence of (i) and (i bis) follows immediately from 4.3, and these conditions plainly imply (ii). The equivalence of (ii) and (ii bis) is also immediate, as is (ii bis) $\implies$ (iii) (cf. 4.1). It remains to prove (iii) $\implies$ (ii bis). This is the only implication that uses the infinitude of k , and it follows at once from the next lemma.

Lemma 4.5. *Let $\mathfrak{d}$ be a nilpotent Lie algebra over an infinite field k , acting on a finite-dimensional vector space V . Suppose that the endomorphism $u(x)$ is not injective for every $x \in \mathfrak{d}$. Then there is a nonzero element $v \in V$ annihilated by $\mathfrak{d}$.*

We may suppose that k is algebraically closed and that $\mathfrak{d}$ is finite-dimensional. Then V is known to be a direct sum of finitely many nonzero stable vector subspaces V_i , $1 \leq i \leq n$, such that, for every i and every $x \in \mathfrak{d}$, the restriction $u(x)|_{V_i}$ has a single eigenvalue $\lambda_i(x)$ (cf. Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Exercise 22). Let $c_i(x)$ be the constant term of the characteristic polynomial of $u(x)|_{V_i}$, so that

$$\lambda_i(x) = 0 \iff c_i(x) = 0.$$

Each c_i is a polynomial function on $\mathfrak{d}$, and the hypothesis says that $\mathfrak{d}$ is the union of their zero sets. Thus one of the c_i is identically zero. Replacing V by V_i , we are reduced to the case in which all $u(x)$, $x \in \mathfrak{d}$, are nilpotent. Engel's theorem (Bourbaki, loc. cit., Th. 1) then implies that some nonzero $v \in V$ is annihilated by $\mathfrak{d}$.

Q.E.D.

It is easy to see, still assuming that k is infinite, that conditions (i) and (i bis) of 4.4 are invariant under every extension of the base field. When they hold, $\mathfrak{d}$ is called a *Cartan subalgebra* of $\mathfrak{g}$. In general, with k not necessarily infinite, $\mathfrak{d}$ is called a Cartan subalgebra of $\mathfrak{g}$ if it becomes one after one, and hence every, extension $k \rightarrow k'$ with k' infinite. It follows in particular that $\mathfrak{d}$ is nilpotent and equal to its own normalizer.

Proposition 4.6. a) *Let $a \in \mathfrak{g}$. If a is regular, then it is contained in one and only one Cartan subalgebra of $\mathfrak{g}$. We shall see the converse in 6.1 d), when k is algebraically closed and $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group.*

b) *Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$, and let $a \in \mathfrak{d}$. Then a is regular in $\mathfrak{g}$ if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.*

For a), if a is regular, then $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$, since this is true after an infinite extension k' of k . It follows at once from 4.3 that every Cartan subalgebra of $\mathfrak{g}$ containing a is equal to this one. For b), observe that the nullity²² of $\text{ad}(a)_{\mathfrak{g}}$ is the sum of the nullities of $\text{ad}(a)_{\mathfrak{d}}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$. Since the first is r , their sum is r if and only if $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective.

Q.E.D.

Corollary 4.7. *Let a be a regular element of $\mathfrak{g}$, let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$ containing a , let A be a k -algebra, and let $\mathfrak{g}_A$ and $\mathfrak{d}_A \subset \mathfrak{g}_A$ be the A -Lie algebras obtained from $\mathfrak{g}$ and $\mathfrak{d}$ by base change; write a_A for the image of a in $\mathfrak{g}_A$. Let u be an automorphism of $\mathfrak{g}_A$. Then*

$$u(\mathfrak{d}_A) = \mathfrak{d}_A$$

if and only if

$$u(a_A) \in \mathfrak{d}_A.$$

The condition is plainly necessary; let us prove that it is sufficient. Suppose it holds. Then

$$\mathfrak{d}' = u^{-1}(\mathfrak{d}_A)$$

is a Lie subalgebra containing a_A , and every $b \in \mathfrak{d}'$ is such that $\text{ad}(b)_{\mathfrak{d}'}$ is nilpotent.²³ Indeed, $\mathfrak{d}'$ is isomorphic to $\mathfrak{d}_A$, which has this property, as follows immediately from the definition of “nilpotent” in Bourbaki, *Groupes et Algèbres de Lie*, Chap. I, §4, Def. 1. Taking $b = a_A$, we see that $\text{Nil}(b, \mathfrak{g}_A)$ contains $\mathfrak{d}'$. On the other hand,

$$\text{Nil}(b, \mathfrak{g}_A) = \mathfrak{d}_A.$$

²²Editor's note: that is, the dimension of the nil-space.

²³The source prints $\mathfrak{d}' = u(\mathfrak{d}_A)$ and $\text{ad}(b)_{\mathfrak{d}}$. Under the hypothesis $u(a_A) \in \mathfrak{d}_A$, the inverse image $u^{-1}(\mathfrak{d}_A)$ is the subalgebra that contains a_A , and the nilpotence assertion concerns its restriction $\text{ad}(b)_{\mathfrak{d}'}$.

Moreover, $\mathfrak{d}'$ is locally a direct summand of $\mathfrak{g}_A$, since $\mathfrak{d}$ is, and hence is locally a direct summand of $\mathfrak{d}_A$. It is a projective module of the same rank r as $\mathfrak{d}_A$, and therefore equals $\mathfrak{d}_A$.

Q.E.D.

Proposition 4.8. *Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$.*

a) *If k is infinite, the following conditions are equivalent:*

- (i) *$\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.*
- (ii) *$\mathfrak{h}$ contains a regular element a of $\mathfrak{g}$, and an element b such that $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ is injective.*
- (iii) *$\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$, and contains a regular element of $\mathfrak{g}$.*

These conditions are invariant under extension of the base field k .

b) *Suppose that these conditions hold after a suitable infinite extension k' of k . For $a \in \mathfrak{h}$, the element a is regular in $\mathfrak{g}$ if and only if it is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective; equivalently, if and only if*

$$\text{Nil}(a, \mathfrak{h}) = \text{Nil}(a, \mathfrak{g})$$

and this common Lie algebra is a Cartan subalgebra of $\mathfrak{h}$.

c) *Under the hypotheses of b), let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. In order that $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{h}$, it is sufficient that it be a Cartan subalgebra of $\mathfrak{g}$. We shall see in 5.8 that this condition is also necessary when $\mathfrak{g}$ is the Lie algebra of a smooth algebraic group G , and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G .*

Conditions (ii) and (iii) of a) are visibly invariant under extension of the base field k , assumed infinite. In b) and c), we may suppose that k is infinite, and shall do so. If $\mathfrak{h}$ contains the Cartan subalgebra

$$\mathfrak{d} = \text{Nil}(a, \mathfrak{g}),$$

then a is a regular element of $\mathfrak{g}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. Thus (i) $\Rightarrow$ (ii). Conversely, if (ii) holds, then a “sufficiently general” element a of $\mathfrak{h}$ simultaneously satisfies both conditions in (ii). Hence $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ contained in $\mathfrak{h}$, proving (i). Conditions (i) and (ii) are therefore equivalent.

Suppose they hold, and let a vary in $\mathfrak{h}$. Then

$$(x) \quad \text{rank}_k \text{Nil}(a, \mathfrak{g}) = \text{rank}_k \text{Nil}(a, \mathfrak{h}) + \text{rank}_k \text{Nil}(\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}).$$

The two terms on the right are respectively at least r' , the nilpotent rank of $\mathfrak{h}$, and at least 0; moreover, both equalities are attained²⁴ for a sufficiently general element of $\mathfrak{h}$. We also have

$$\text{rank}_k \text{Nil}(a, \mathfrak{g}) \geq r,$$

where r is the nilpotent rank of $\mathfrak{g}$. Equality is attained for a sufficiently general element of $\mathfrak{h}$, and holds if and only if a is regular in $\mathfrak{g}$.

It follows that $r = r'$, and that a is regular if and only if the two terms on the right-hand side of (x) are respectively r' and 0; that is, if and only if a is regular in $\mathfrak{h}$ and $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective. This proves b), and c) follows immediately by taking an element $a \in \mathfrak{d}$ regular in $\mathfrak{g}$, so that $\text{Nil}(a, \mathfrak{g}) = \mathfrak{d}$.

The preceding result also proves (i) $\Rightarrow$ (iii). Finally, (iii) $\Rightarrow$ (i): under (iii), a sufficiently general $a \in \mathfrak{h}$ is regular both in $\mathfrak{h}$ and in $\mathfrak{g}$. Thus

$$\text{Nil}(a, \mathfrak{h}) \subset \text{Nil}(a, \mathfrak{g})$$

²⁴Editor’s note: “the equality” has been replaced by “the equalities”.

are respectively Cartan subalgebras of $\mathfrak{h}$ and $\mathfrak{g}$. Since they have the same rank over k , they are equal, proving (i). This completes the proof of 4.8.

5 The case of the Lie algebra of a smooth algebraic group: the density theorem

Let G be a smooth algebraic group over the field k , and let $\mathfrak{g}$ be its Lie algebra. Let $\mathfrak{h}$ be a Lie subalgebra of $\mathfrak{g}$. Let G act on $W(\mathfrak{g})$ through the adjoint representation, and consider the subscheme $W(\mathfrak{h})$. The construction of Section 1 leads us to introduce

$$N = \text{Norm}_G(\mathfrak{h}) = \text{Norm}_G(W(\mathfrak{h})),$$

which is an algebraic subgroup of G (not necessarily smooth), and

$$\mathfrak{n} = \text{Lie}(N),$$

together with the scheme

$$X = G \times^N W(\mathfrak{h}),$$

the quotient of $\tilde{X} = G \times W(\mathfrak{h})$ by the right action of N given by

$$(g, x) \cdot n = (gn, \text{Ad}(n^{-1})x).$$

We consider the canonical morphisms

$$\begin{array}{ccc} G \times W(\mathfrak{h}) & & \\ q \downarrow & \searrow \varphi & \\ X & \xrightarrow{\psi} & W(\mathfrak{g}). \end{array}$$

Theorem 5.1. *With the preceding notation, suppose that k is infinite. Consider the following conditions:*

- (i) $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (ii) There exists $a \in \mathfrak{h}$ such that $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective.
- (iii) The morphism $\varphi : G \times W(\mathfrak{h}) \rightarrow W(\mathfrak{g})$ is generically smooth.
- (iv) The preceding morphism φ (or, equivalently, $\psi : X \rightarrow W(\mathfrak{g})$) is dominant, and $\mathfrak{h}$ has the same nilpotent rank as $\mathfrak{g}$.
- (v) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is generically smooth and $\mathfrak{h} = \mathfrak{n}$.
- (vi) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and $\mathfrak{h} = \mathfrak{n}$.
- (vii) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is dominant.
- (viii) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, and N is smooth.
- (ix) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is dominant, $\mathfrak{h} = \mathfrak{n}$, and $\mathfrak{h}$ is the Lie algebra of a smooth algebraic subgroup H of G .
- (x) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is generically quasi-finite, and $\mathfrak{n} = \mathfrak{h}$.

- (xi) The morphism $\psi : X \rightarrow W(\mathfrak{g})$ is generically étale, and $\mathfrak{n} = \mathfrak{h}$.
- (xii) There exists a smooth algebraic subgroup H of G , with Lie algebra $\mathfrak{h}$, such that $\mathfrak{h}$ contains a Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$.
- (xiii) There exists $a \in \mathfrak{h}$ such that N and the transporter M_a of a into $\mathfrak{h}$ (cf. Section 1) coincide in a neighborhood of e , and $\mathfrak{n} = \mathfrak{h}$.

One then has the following diagram of implications:

$$\begin{array}{ccccccccccc}
 \text{(xi)} & \longleftrightarrow & \text{(xii)} & \longleftrightarrow & \text{(xiii)} & \implies & \text{(viii)} & \longleftrightarrow & \text{(ix)} & \longleftrightarrow & \text{(x)} \\
 & & \Downarrow & & & & & & \Downarrow & & \\
 \text{(i)} & \longleftrightarrow & \text{(ii)} & \longleftrightarrow & \text{(iii)} & \longleftrightarrow & \text{(iv)} & \implies & \text{(v)} & \implies & \text{(vi)} \implies \text{(vii)}.
 \end{array}$$

When k has characteristic zero, all the conditions under consideration are equivalent. Finally,

$$\text{(xi)} \longleftrightarrow [(\text{i}) \text{ and } (\text{viii})] \longleftrightarrow [(\text{v}) \text{ and } (\text{viii})].$$

Let us first record the trivial implications

$$(\text{v}) \implies (\text{vi}) \implies (\text{vii}), \quad (\text{ix}) \implies (\text{vi}), \quad (\text{xi}) \longleftrightarrow [(\text{x}) \text{ and } (\text{v})].$$

We prove the equivalence of conditions (i) through (iv), and that they imply (v). The implication (i) $\implies$ (ii) is immediate. On the other hand, when k is algebraically closed, condition (iii) says that there is a k -rational point of $G \times W(\mathfrak{h})$ at which the tangent map of φ is surjective. It is clear that, after translating by a suitable element of $G(k)$, this point may be taken to have the form (e, a) , with $a \in \mathfrak{h}$. It follows that when k is infinite, though not necessarily algebraically closed, this condition—which is plainly sufficient for generic smoothness—is also necessary.

The tangent map is readily computed. Upon identifying the tangent space of $W(\mathfrak{h})$ at a with $\mathfrak{h}$, it is the map

$$(\xi, x) \longmapsto [\xi, a] + x$$

from $\mathfrak{g} \times \mathfrak{h}$ to $\mathfrak{g}$. Its surjectivity is equivalent to the surjectivity of $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$, or, what amounts to the same thing, to its injectivity. This proves the equivalence of (ii) and (iii). Moreover, (ii) plainly implies $\mathfrak{h} = \mathfrak{n}$, and (iii) implies that ψ is generically smooth: if φ is smooth at a point u , then, since q is flat, ψ is smooth at $q(u)$. Thus (ii) and (iii) imply (v).

We now show that they imply (i). Since ψ is dominant and the set of regular points of $\mathfrak{g}$ is dense and open, $\mathfrak{h}$ contains regular elements of $\mathfrak{g}$. Hence a sufficiently general element b of $\mathfrak{h}$ is regular in $\mathfrak{g}$ and makes $\text{ad}(b)_{\mathfrak{g}/\mathfrak{h}}$ injective. Consequently $\text{Nil}(b, \mathfrak{g}) \subset \mathfrak{h}$, so $\mathfrak{h}$ contains the Cartan subalgebra $\mathfrak{d} = \text{Nil}(b, \mathfrak{g})$. Conditions (i), (ii), and (iii) are therefore equivalent. Finally, (i) $\iff$ (iv): we have already observed that if $\mathfrak{h}$ contains a Cartan subalgebra, then it has the same nilpotent rank as $\mathfrak{g}$ (4.6), so (i) implies (iv). Conversely, if (iv) holds, then $\mathfrak{h}$ contains a regular element of $\mathfrak{g}$, and since it has the same rank as $\mathfrak{g}$, it contains a Cartan subalgebra by 4.6.

We next prove the equivalence of conditions (viii) through (x). We first note the following facts.

Lemma 5.2. *a) If N is smooth, then*

$$\dim X \leq \dim G,$$

with equality if and only if $\mathfrak{h} = \mathfrak{n}$.

b) If $\mathfrak{h} = \mathfrak{n}$, then

$$\dim X \geq \dim G,$$

with equality if and only if N is smooth.

These assertions follow at once from the formula

$$\dim X = \dim G - \dim N + \dim_k \mathfrak{h}$$

and from the fact that $\dim N = \dim_k \mathfrak{n}$ is equivalent to the smoothness of N .

Now (viii) implies (x): by dominance, (viii) gives $\dim X \geq \dim G$, so Lemma 5.2(a) gives equality and $\mathfrak{h} = \mathfrak{n}$, hence (x). The same argument in reverse, using Lemma 5.2(b), shows that (x) implies (viii). Moreover, (viii) implies (ix), since it implies $\mathfrak{h} = \mathfrak{n}$, so $\mathfrak{h}$ is the Lie algebra of the smooth algebraic subgroup N of G . Conversely, (ix) implies (viii): since H normalizes its Lie algebra $\mathfrak{h}$, it is contained in N ; and since H is smooth and has the same Lie algebra as N , it follows that N is smooth.

Finally, we prove the equivalence of (xi), (xii), and (xiii), and show that they imply (iii); this will complete the implication diagram. We have (xi) $\iff$ (xiii). Indeed, if $\mathfrak{n} = \mathfrak{h}$, then Lemma 5.2(b) gives $\dim X \geq \dim G$. Since $W(\mathfrak{g})$ is normal, condition (xi) is therefore equivalent to ψ being generically unramified, and this in turn is equivalent to (xiii) by 1.1, (ii) $\iff$ (iii), arguing as above in the proof of (ii) $\iff$ (iii).

Since (xi) $\Rightarrow$ (x) $\Rightarrow$ (viii), what we have already proved shows that (xi) implies that N is smooth. Thus $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth, and the composite $\varphi = \psi \circ q$ is generically smooth; this is (iii). Since (iii) implies (i), it also follows that (xi) $\Rightarrow$ (xii).²⁵ Finally, (xii) implies (xi). Indeed, (xii) plainly implies (i), and we have seen that (i) $\Rightarrow$ (iii) $\Rightarrow$ (v); hence (xii) implies (v). It also implies (ix), and (ix) implies (x). Thus (xii) implies both (v) and (x), and therefore (xi), since

$$\text{generically étale} = \text{generically smooth} + \text{generically quasi-finite}.$$

If k has characteristic zero, then (vii) implies (viii), because Cartier's theorem makes N automatically smooth (VI_B 1.6.1). Moreover, [(viii) and (x)] implies (xi), because in characteristic zero, for a morphism of integral preschemes,

$$\text{generically étale} = \text{dominant and generically quasi-finite}.$$

Thus in this case all conditions (i) through (xiii) are equivalent.

Corollary 5.3. *Under the equivalent conditions (viii) through (x), there exists a unique smooth connected algebraic subgroup H of G whose Lie algebra is $\mathfrak{h}$, and*

$$\text{Norm}_G(H) = \text{Norm}_G(\mathfrak{h}) = N, \quad H = N^0.$$

Indeed, $H = N^0$ has the required properties. Conversely, if H has these properties, then $H \subset N$, since H normalizes its Lie algebra $\mathfrak{h}$. This is an inclusion of smooth groups with the same Lie algebra, and H is connected, so $H = N^0$. To prove $\text{Norm}_G(H) =$

²⁵Editor's note: because $H = N$ will do.

$\text{Norm}_G(\mathfrak{h})$, we may assume that k is algebraically closed. What we have just proved then shows immediately that the two groups have the same k -points. On the other hand, the inclusions

$$H \subset \text{Norm}_G(H) \subset N$$

show that $\text{Norm}_G(H)$ and N have the same Lie algebra; hence they are equal.

Corollary 5.4. *Under the equivalent conditions (i) through (iv), let $a \in \mathfrak{h}$. Then the following conditions are equivalent, and they hold whenever a is regular in $\mathfrak{g}$:*

(i) φ is smooth at (e, a) .

(ii) $M_a = \text{Transp}_G(a, \mathfrak{h})$ is smooth at e , and $\dim_e(M_a) = \text{rank}_k \mathfrak{h}$.

(iii) $\text{ad}(a)_{\mathfrak{g}/\mathfrak{h}}$ is injective (equivalently, bijective).

Under the equivalent conditions (xi) through (xiii), let H be the algebraic subgroup of G considered in Corollary 5.3. The preceding conditions are also equivalent to:

(iv) ψ is étale at $(\bar{e}, a)$.

(v) If M_a^0 denotes the connected component of e in the transporter M_a of a into $\mathfrak{h}$, endowed with the structure induced by M_a , then

$$H = M_a^0.$$

Plainly (i) implies (ii), since M_a is isomorphic to the fiber $\varphi^{-1}(a)$, with e corresponding to (e, a) . Conversely, (ii) implies (i): condition (ii) says that φ is “equidimensional” at (e, a) , in the sense that the dimension of the fiber through this point is the dimension of the generic fiber. Since $G \times W(\mathfrak{h})$ and $W(\mathfrak{g})$ are regular, this implies that φ is flat at (e, a) , hence smooth there because its fiber is smooth at that point. The equivalence of (i) and (iii) was obtained in the proof of Theorem 5.1 by the elementary calculation of the tangent map. We also saw in 4.8(b) that regularity of a in $\mathfrak{g}$ implies (iii).

Under conditions (xi) through (xiii), the morphism $q : G \times W(\mathfrak{h}) \rightarrow X$ is smooth, since N is smooth. Thus (i) is equivalent to the smoothness of ψ at $(\bar{e}, a)$; and since ψ is generically étale, this is equivalent to (iv). Finally, as was noted at the end of the proof of 1.1, condition (iv) implies that N is the prescheme induced on M_a by an open and closed subset of M_a , which gives (v). Conversely, (v) trivially implies (ii). Alternatively, (v) implies (iv) by 1.1, since ψ is dominant and $W(\mathfrak{g})$ is normal, and “unramified” is equivalent here to “étale”. This completes the proof of Corollary 5.4.

Corollary 5.5. *Let G be a smooth algebraic group over a field k , and let H be a smooth algebraic subgroup such that its Lie algebra $\mathfrak{h}$ contains, at least after a suitable extension of k , a Cartan subalgebra of the Lie algebra $\mathfrak{g}$ of G . Let K be a connected algebraic subgroup of G , not necessarily smooth, with Lie algebra $\mathfrak{k}$, and suppose that $\mathfrak{k}$ contains a regular element a of $\mathfrak{g}$, again at least after a suitable extension of k . Then H contains K if and only if $\mathfrak{h}$ contains $\mathfrak{k}$.*

Indeed, Corollary 5.4, (iii) $\Rightarrow$ (v), applied to the connected subgroup supplied by Corollary 5.3, gives $H^0 = M_a^0$.²⁶ This identity is invariant under extension of the base field, so it remains valid without assuming that the base field is infinite. Moreover, $\mathfrak{k} \subset \mathfrak{h}$ plainly implies $K \subset M_a$. Since K is connected, $K \subset M_a^0 \subset H$. $\square$

²⁶The French source prints $H = M_a^0$ here. In the stated generality the argument gives $H^0 = M_a^0$, which is all that is needed.

Corollary 5.6. *Let G and H be as in Corollary 5.5, and let K be an algebraic subgroup of G . Suppose that H is connected and K is smooth. Then K contains H if and only if $\mathfrak{k}$ contains $\mathfrak{h}$.*

Indeed, if $\mathfrak{k} \supset \mathfrak{h}$, then K satisfies, with respect to H , the hypothesis imposed on H in Corollary 5.5. Meanwhile H plainly satisfies, with respect to K , the condition imposed on K there. The conclusion follows from Corollary 5.5.

Corollary 5.7. *Let $\mathfrak{g}$ be the Lie algebra of a smooth algebraic group G over a field k . Then:*

- a) A Lie subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ is a Cartan subalgebra if and only if it is nilpotent and equal to its own normalizer.*
- b) An element $a \in \mathfrak{g}$ is regular if and only if $\text{Nil}(a, \mathfrak{g})$ is nilpotent.*

After extending the base field, we may assume that k is infinite. In view of 4.4, to prove (a) it remains only to show that if $\mathfrak{d}$ is nilpotent and contains an element a for which $\text{ad}(a)_{\mathfrak{g}/\mathfrak{d}}$ is injective, then $\mathfrak{d}$ is a Cartan subalgebra. By Theorem 5.1, (ii) $\Rightarrow$ (i), $\mathfrak{d}$ contains a Cartan subalgebra

$$\mathfrak{d}' = \text{Nil}(a, \mathfrak{g}).$$

Since $\mathfrak{d}$ is nilpotent, 4.3 now gives $\mathfrak{d} = \mathfrak{d}'$. For (b), one notes that $\text{Nil}(a, \mathfrak{g})$ is a Cartan subalgebra of $\mathfrak{g}$ by (a), and therefore a is regular.

Corollary 5.8. *Let G be a smooth algebraic group over a field k , and let H be a smooth algebraic subgroup, with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Suppose that after a suitable extension of the base field, $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$. Let $\mathfrak{d}$ be a Lie subalgebra of $\mathfrak{h}$. Then $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$ if and only if it is a Cartan subalgebra of $\mathfrak{g}$.*

In view of 4.8(c), it remains to show that if $\mathfrak{d}$ is a Cartan subalgebra of $\mathfrak{h}$, then it is one of $\mathfrak{g}$. It is enough to show that $\mathfrak{d}$ contains an element a regular in $\mathfrak{g}$, and we may assume that k is algebraically closed. But there is a dense open subset of $\mathfrak{h}$ consisting of points regular in $\mathfrak{g}$, so the assertion follows from Theorem 5.1, (i) $\Rightarrow$ (vii), applied to $(\mathfrak{h}, \mathfrak{d})$.

6 Cartan subalgebras and subgroups of type (C) associated with a smooth algebraic group

For simplicity, in the following theorem we restrict ourselves to an algebraically closed base field; the case of an arbitrary base prescheme will be treated in the next Exposé.

Theorem 6.1. *Let G be a smooth algebraic group over an algebraically closed field k , and let $\mathfrak{g}$ be its Lie algebra. Then:*

- a) The Cartan subalgebras of $\mathfrak{g}$ are conjugate.*
- b) Let $\mathfrak{d}$ be a Cartan subalgebra of $\mathfrak{g}$. Its normalizer N in G is smooth, and $D = N^0$ is the unique smooth connected subgroup of G whose Lie algebra is $\mathfrak{d}$. One has*

$$\text{Norm}_G(\mathfrak{d}) = \text{Norm}_G(D) = N, \quad \text{and hence} \quad D = \text{Norm}_G(D)^0.$$

c) With $\mathfrak{d}$ as in (b), set, as in Section 5,

$$X = G \times^N W(\mathfrak{d}),$$

and consider the canonical morphism

$$\psi : X \longrightarrow W(\mathfrak{g}).$$

Its fiber over $a \in \mathfrak{g}$ has as its k -valued points the set of Cartan subalgebras of $\mathfrak{g}$ containing a . Then ψ is birational.

d) With the notation of (c), let U be the largest open subset of $W(\mathfrak{g})$ for which ψ induces an isomorphism

$$\psi^{-1}(U) \xrightarrow{\sim} U.$$

Then, for $a \in \mathfrak{g}$, the following conditions are equivalent:

- (i) $a \in U(k)$.
- (ii) The element a belongs to one and only one Cartan subalgebra of $\mathfrak{g}$.
- (iii) The set of Cartan subalgebras of $\mathfrak{g}$ containing a is finite and nonempty.
- (iv) The fiber $\psi^{-1}(a)$ has an isolated point.
- (v) If $a \in \mathfrak{d}$, the morphism ψ is étale—or merely quasi-finite—at the point $(\bar{e}, a)$.
- (vi) The element a is regular in $\mathfrak{g}$.
- (vii) With the notation of Corollary 5.4(iii), one has $N = M_a$.

Proof. Apply Theorem 5.1 with the Cartan subalgebra $\mathfrak{d}$ of $\mathfrak{g}$ in place of $\mathfrak{h}$. We show that the strongest conditions, (xi) through (xiii), hold. This is clear either in the form (xiii), by 4.7—which says that if a regular element $a \in \mathfrak{g}$ lies in $\mathfrak{d}$, then $N = M_a$ —or in the form

$$(xi) = (x) + (i).$$

Indeed, (i) is automatic, while (x) follows from the fact that a regular point of $\mathfrak{g}$ belongs to a unique Cartan subalgebra of $\mathfrak{g}$ (4.6(a)), and therefore, a fortiori, to a unique conjugate of $\mathfrak{d}$.

Part (b) now follows from Corollary 5.3. Part (c) follows because ψ is generically étale and because a sufficiently general point of $\mathfrak{g}$ —more precisely, a regular point—belongs to a unique Cartan subalgebra of $\mathfrak{g}$. Under these conditions, the equivalence of conditions (i) through (v) on a follows immediately from Zariski's Main Theorem, applied to the separated birational morphism ψ , since $W(\mathfrak{g})$ is normal and X is integral. The equivalence of (v) and (vi) is the special case of Corollary 5.4, (iii) $\iff$ (iv), obtained after carrying a into $\mathfrak{d}$ by a suitable element $g \in G(k)$, in view of 4.6(b). Moreover, Corollary 5.4 shows that (v) and (vi) are also equivalent to

$$N \supset M_a^0.$$

By 4.7 this even implies $N = M_a$.²⁷ This proves (d).

Of course, (b), (c), and the equivalence of (i), (iv), (v), (vi), and (vii) remain valid over an arbitrary base field. We now prove (a), using the assumption that k is algebraically closed. By Theorem 5.1, (i) $\implies$ (vii), the morphism $\psi : X \rightarrow W(\mathfrak{g})$ is dominant. Hence

²⁷The French source prints $N = M$ at this point; condition (vii) and the surrounding notation require $N = M_a$.

there is a dense open subset V of $W(\mathfrak{g})$ such that every $a \in V(k)$ is the image of a point of $X(k)$, or equivalently, belongs to a conjugate of $\mathfrak{d}$. Apply the same result to a second Cartan subalgebra $\mathfrak{d}'$ of $\mathfrak{g}$. We may choose V so that every element of $V(k)$ is conjugate both to an element of $\mathfrak{d}$ and to an element of $\mathfrak{d}'$. Choose a regular element of $V(k)$. Some conjugate $\mathfrak{d}''$ of $\mathfrak{d}'$ then contains an element of $\mathfrak{d}$ regular in $\mathfrak{g}$, and 4.6(a) gives $\mathfrak{d}'' = \mathfrak{d}$. This proves (a) and completes the proof. $\square$

Definition 6.2. Let G be a smooth algebraic group over a field k . A *subgroup of type (C)* of G is a smooth connected algebraic subgroup whose Lie algebra is a Cartan subalgebra of $\mathfrak{g}$. The *infinitesimal rank* of G is the nilpotent rank of its Lie algebra $\mathfrak{g}$; it is equal to the dimension of every subgroup of type (C) of G .

Theorem 6.1(b) immediately gives:

Corollary 6.3. *Under the hypotheses of Definition 6.2, the map*

$$D \longmapsto \mathfrak{d} = \text{Lie}(D)$$

establishes a bijection between the subgroups of type (C) of G and the Cartan subalgebras $\mathfrak{d}$ of $\mathfrak{g}$. If D is a subgroup of type (C) of G , then D is its own connected normalizer:

$$D = \text{Norm}_G(D)^0.$$

Combining Theorem 6.1(a) and Corollary 6.3 gives:

Corollary 6.4. *If k is algebraically closed, the subgroups of type (C) of G are conjugate to one another.*

Corollary 6.5. *Let G be a smooth algebraic group over an algebraically closed field k , and let H be a smooth algebraic subgroup of G , with Lie algebras $\mathfrak{g}$ and $\mathfrak{h}$, respectively. Then $\mathfrak{h}$ contains a Cartan subalgebra of $\mathfrak{g}$ if and only if H contains a subgroup D of type (C) of G .*

The condition is plainly sufficient. It is also necessary, since by Corollary 5.5,

$$H \supset D \iff \mathfrak{h} \supset \mathfrak{d}.$$

Remarks 6.6.

- a) The connected subgroups H of G described in Corollary 6.5 correspond bijectively to the Lie subalgebras $\mathfrak{h}$ of $\mathfrak{g}$ satisfying the strongest condition, (xii), of Theorem 5.1. By Corollary 5.5, inclusion relations among such subgroups can already be read from their Lie algebras.
- b) Continue to suppose that k is algebraically closed, and let D be a subgroup of type (C) of G . It is then easy to show that D contains a Cartan subgroup C of G . Indeed, put $V = \mathfrak{g}/\mathfrak{d}$. For a general element $a \in \mathfrak{d}$, the endomorphism $\text{ad}(a)$ of V is injective; it suffices to take a regular in $\mathfrak{g}$. From this one readily deduces that, for a general $g \in D(k)$, the action of $\text{Ad}(g)$ on V has no nonzero invariant vector. Thus 2.1, (vii) $\Rightarrow$ (i), applies. In fact, in the next Exposé we shall prove the more precise result that every Cartan subgroup C of G is contained in one and only one subgroup of type (C) of G .

- c) In general one must not confuse Cartan subgroups with subgroups of type (C) . The subgroups of type (C) of G are Cartan subgroups if and only if they are nilpotent, since Cartan subgroups are maximal nilpotent subgroups. It may happen that $\mathfrak{g}$ is nilpotent although G is not; an example is $\mathrm{SL}(2)_k$ when k has characteristic 2. Then the Cartan subalgebras of $\mathfrak{g}$ are all equal to $\mathfrak{g}$. Thus, if G is connected, G is a subgroup of type (C) of itself, but it is not a Cartan subgroup of itself.

On the other hand, we shall see in Exposé XIV that if G is an adjoint semisimple group, its subgroups of type (C) are precisely its Cartan subgroups. The same conclusion holds in characteristic zero for every smooth algebraic group G over k . This follows at once from the fact that, in characteristic zero, a connected algebraic group is nilpotent if and only if its Lie algebra is nilpotent. In turn, this follows from the fact that in characteristic zero the center Z of a connected algebraic group G has Lie algebra equal to the center of $\mathfrak{g}$. Indeed, one obtains a priori the space of invariants in $\mathfrak{g}$ under the adjoint representation of G . Since G is connected and k has characteristic zero, these are also the infinitesimal invariants of the adjoint representation of $\mathfrak{g}$, namely the center of $\mathfrak{g}$. Finally, Cartier's theorem (cf. VI_B 1.6.1) shows that Z is smooth.